The effect of alkalinity enrichment on coral larval settlement

September 2025 pilot trial

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1 Introduction

The success and continued feasibility of coral restoration efforts necessitates they scale to meet the magnitude of the ecological crisis. Addressing this challenge requires approaches that accelerate the production of coral outplants while reducing cost and effort. Although hobby aquarists commonly enrich seawater alkalinity to stimulate coral growth, this practice has rarely been examined in the scientific literature and many questions remain regarding effectiveness on species targeted for restoration and at early life stages.

Restoration practitioners are increasingly prioritizing the sexual production of corals, which enables the deliberate crossing of desirable genotypes, enhances genetic diversity, and supports the creation of more resilient outplants. This research examines whether alkalinity enrichment can be integrated into this process, with a particular focus on its effects on larval settlement and early post-settlement development.

Gametes from two Caribbean coral species, *Orbicella faveolata* (Ofav) and *Pseudodiploria strigosa* (Pstr), both commonly targeted for restoration, were collected offshore of Key Largo on 15 August 2025 and transported to the Rosenstiel School of Marine and Atmospheric Science on Virginia Key for fertilization (Figure 1). Once larvae reached competency, they were transferred into 8-oz jars assigned to four treatments (ambient control and three elevated-alkalinity levels). Larvae were allowed to settle on ceramic tiles, after which settlement was quantified.

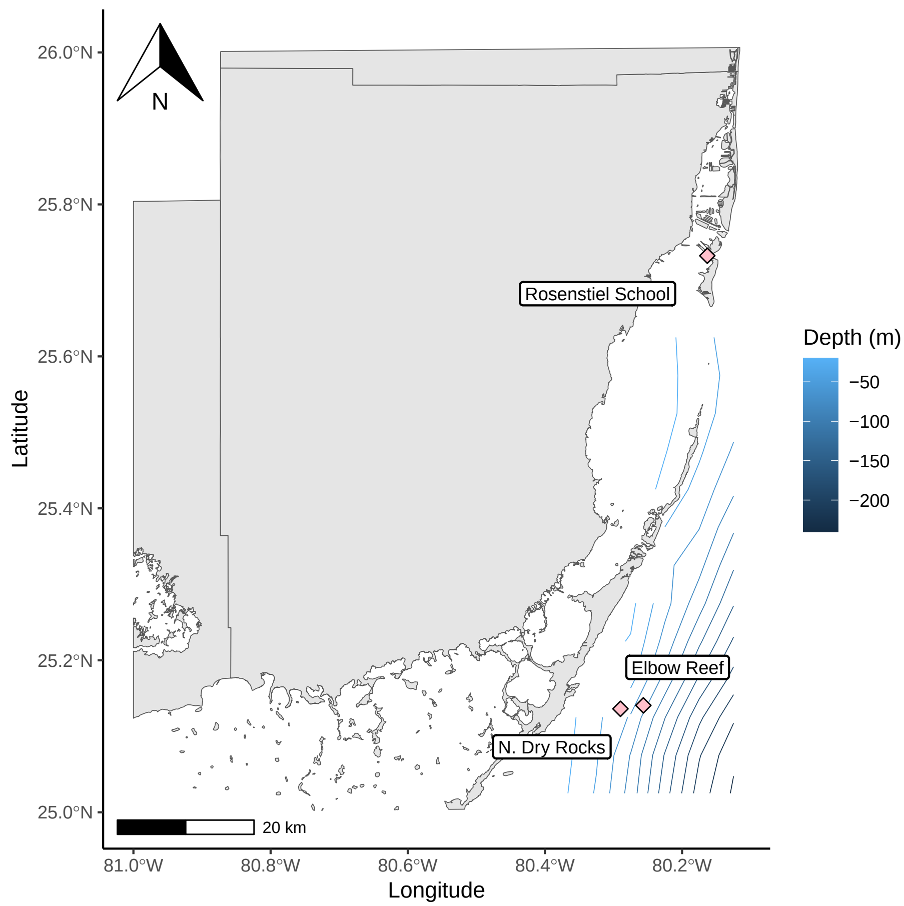

Figure 1: Coral gametes were collected from reefs off of Key Largo, then transported and fertilized at the Rosenstiel School.

2 Objectives

1. Evaluate the feasibility and stability of water-chemistry manipulation for alkalinity enrichment.
2. Assess the ability to induce and quantify larval settlement under controlled laboratory conditions.
3. Determine settlement success across multiple alkalinity treatments.
4. Identify and resolve practical challenges in the experimental setup, including dosing, monitoring, and replication logistics.

3 Data sources

Alkalinity water-chemistry data

Water samples were analyzed using a Metrohm robotic alkalinity titrator, and raw data were exported directly from the Tiamo software. Datasets include:

1. Sample metadata (generated by KMC):
    - "bottle.csv"
2. Raw output from Tiamo software:
    - "settlement_alk_data.csv"

Coral metadata and settlement counts

1. Settlement data:
    - "settlement_data.csv"

Map figure data

1. Florida county geopackage:
    - https://gis-fdot.opendata.arcgis.com/datasets/florida-county-boundaries-with-fdot-districts/explore
2. Reef coordinates:
    - "reefs" tribble

```
# A tibble: 3 x 4
  reef                 lat   lon species
  <chr>              <dbl> <dbl> <chr>
1 N. Dry Rocks        25.1 -80.3 Orbicella faveolata
2 Elbow Reef          25.1 -80.3 Pseudodiploria strigosa
3 Rosenstiel School   25.7 -80.2 NA
```

4 Data processing

Water chemistry data

The raw output from Tiamo software, "settlement_alk_data.csv", was first cleanaed and processed. The sample_id field was standardized by trimming with str_extract() to retain only the bottle number, allowing it to be accurately joined to the bottle metadata using left_join(). Because each sample had two replicate total alkalinity (TA) measurements, one final TA and standard deviation value were created by averaging the replicates and using mutate() to generate a new column. The final processed dataset was exported as an .rds and retained only the cleaned bottle ID (sample_num), final TA (ta_av), and TA standard deviation (ta_sd).

The processed output was then joined to the bottle metadata. Treatment statistics were generated using group_by() and summarize().

Coral settlement data

To analyze the settlement data, the raw data was filtered to exclude vessels which did not contain corals. Treatment statistics for total number of settlers across vessels by treatment were calculated with group_by() and summarize(). A new collumn for percent settlement per vessel was calculated and created with mutate(). Plot created with stat_summary() in ggplot.

5 Main findings

5.1 Alkalinity treatment summary

The four treatments were well grouped and on target (Figure 2).

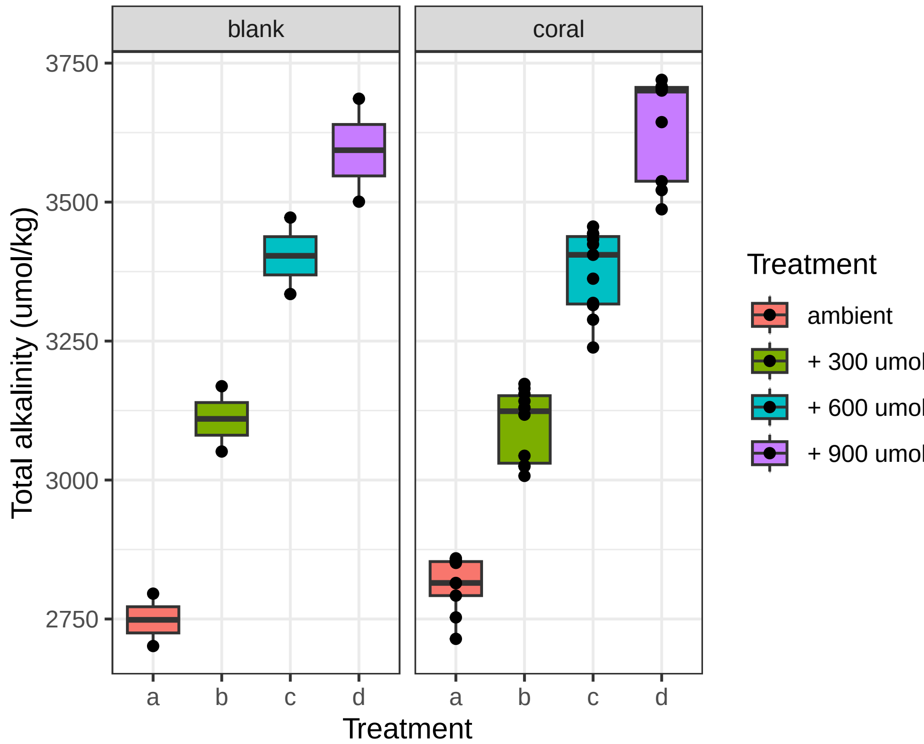

Figure 2: Overall treatment summaries including all water samples collected over the course of the experiment.

This pilot used artificial seawater which was mixed once at the beginning of the experiment into two large vats, serving as source water for treatment a and treatment d. Intermediate treatments b and c were created by mixing water from the vats in specific ratios. The calculated dosing ratios largely held the intended 300 µmol increase, sustaining clear separation between treatment levels. Difference in µmols between treatments averaged out to:

a and b: +326.63

b and c: +294.07

c and d: +248.62

The mean alkalinity of treatment "a", meant to represent ambient seawater, was higher than a typical seawater alkalinity, 2713.07 versus ~2400 umol/kg. This was due to the background alkalinity of InstantOcean salt being variable between batches. Will use natural seawater as the base in the future. Full treatment summaries with standard deviation in Table below.

treatment	mean	pH	sd	n
a	2713.066	7.857330	162.05025	15
b	3039.696	7.901942	140.24409	15
c	3333.767	7.945160	119.71223	16
d	3582.387	7.981334	124.91603	14
source_a	2625.623	7.976561	44.92218	2
source_d	3387.669	7.998353	108.95018	2

5.2 Treatment stability over time

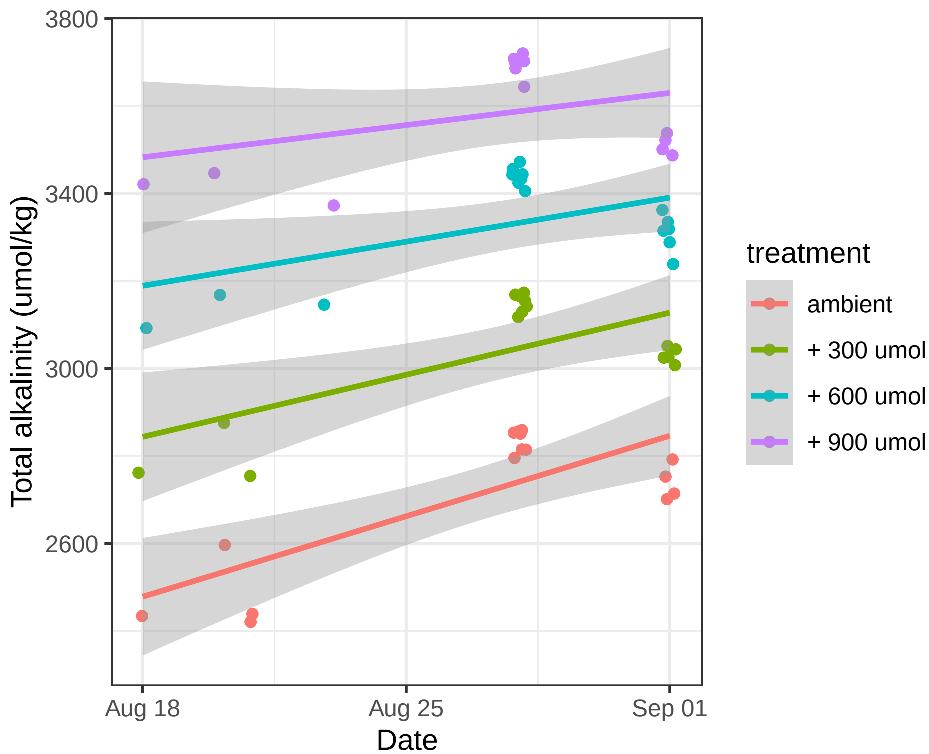

Figure 3: Alkalinity treatment stability over time with liner regression.

Linear regression shows an increase in alkalinity across all treatments over time, likely due to issues of evaporation in the intermediate sump and from the vessels leading to higher salinity and TA. Will address this in subsequent experiments by increasing turnover rate within vessels and by using flow-through natural seawater system instead of artificial seawater reservoirs.

5.3 Larval settlement

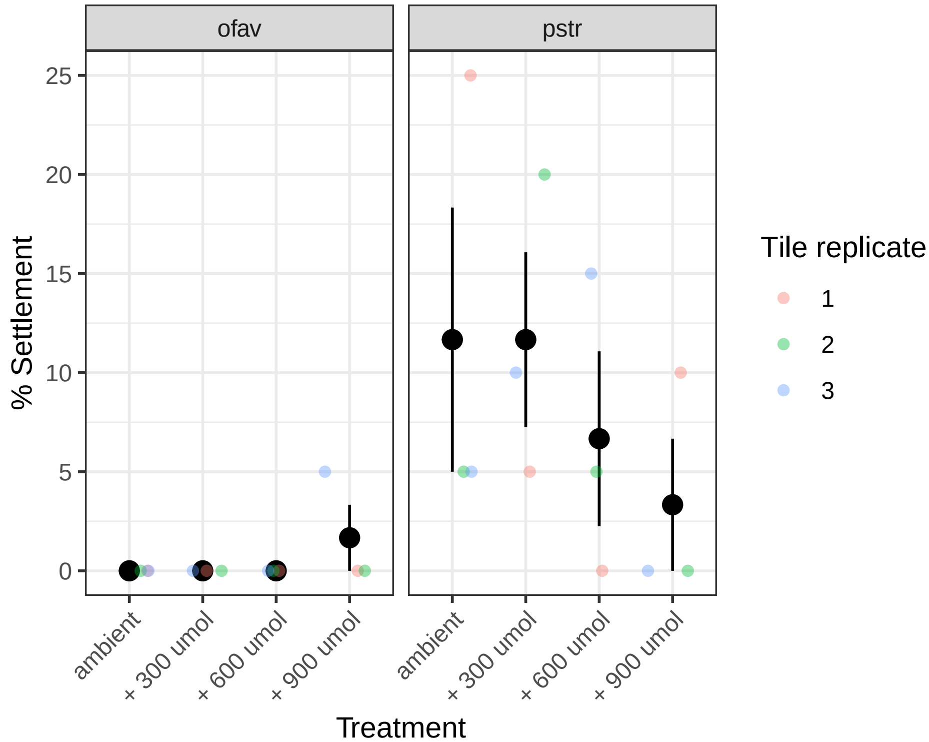

Figure 4: Percent of larvae settled out of 20 per tile, faceted by species. Each treatment included three replicates.

Settlement was overall lower than expected for both species. Had little to no settlement of Ofav larvae likely due to a spike in salinity. Issues of ambient lighting in the lab might have also impacted settlement rates.